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Day 2

SDLC (Software Development lifecycle)

It is a structured process used from an initial idea or requirement to development testing, release, and eventual maintenance or replacement the whole complete journey of the software.

Why need SDLC?

Without a defined development process, a software project can become difficult to manage.

For eg: imagin building a student reg app without deciding
- what the app should do
- who will use it
- what req it most satisfy
- How it's system will work / tech stack

We may endup with
- missing requirements
- poor design
- Unexpected defects
- Rework, Delayed
- Increased cost
- Poor user experience.

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Phases of SDLC

i) Requirement gathering

In this phase of of SDLC the team gather requirement from stakeholders, developer, adviser or expert such as customers, end user, business teams, management, .

eg:- for student reg app some of the req might be

- student must be able to login / create new acc.
- One student is not allowed to creat mult acc
- system show all available seats.

QA can reiview the requirements and ask

- is the requirement clear
- Are any condition are missing.
- what will happen if reg fail.

ii) Design

In this phase the team decides how the system will work. what a will need like system architecture, Database design, UI desig, Data flow, etc.

QA mainly review the design for testability, missing error handling, security risk, and overall designs of the project.

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iii) Development / Implement.

Developers convert the design and req into a fully working software. Activities includes writing code, creating database table, Developing API Implementing business logic

QA may prepare test case write code review test and Review acceptance criteria

iv) Testing

It determinines whether the software works as expected and satisfies the requirements. It consis. of multi testing such as unit, integration, system Acceptance

v) Deployment

Once the software has passed the required checks or Acceptance test ~~then~~ it is approved for release, it is deployed to target env. In include configure server, setting environment variables, confi networking.

QA run smoke test while monitoring the releases.

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vi) Maintenance

The deployment doesnot means the end of the software ~~e~~ even after the completion we may have to fix defects, Improve performance, Add features, Patch security vulnerabilities, update dependencies, monitor production, support users.

• Sequential Models

These SDLC model follows a specific path with per-defined phases happen one after another (requirement → design → build → test → release), like a stair case but you cannot go back.

Waterfall Model

• Waterfall model is a sequential SDLC model where software development move through each phase step by step - Each phase is generally completed before the next one begins

• Waterfall Model ~~has~~ follows a fixed order
• The requiriments are defined before development
• Main testing happens ~~before~~ after dev
• Late changes ~~d~~ cost more resourses

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eg:- Student Registration App

- Requirement: student can register for courses
- Design: Design the reg. page and bd
- Development: Developer build the sys
- Testing: QA test login, reg, course capacity.
- Deployment: The completed app is released to students.

V- Model

It's similer to waterfall model but with more emphasizes on verification and validation. Each development phase has a corresponding testing phase

Requirement Analysis ← Acceptance Test Design → Acceptance Testing

↓

System Design ← System Test Design → System Testing

↓

Architecture Design ← Integration Test Design → Integration Testing

↓

Moduler Design ← Unit Test design → Unit Testing

→ Coding

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- Testing starts from early stage
- Each level have it's own test level

For eg: Banking ATM System

- Requirements - ATM should allow users to withdraw money → Acceptance Testing
- System Design: Design the complet ATM system → System Testing
- Module Design: Design ATM, bank server, and database interactions → Integration Testing
- Coding: Develop withdrawal and balance-checking function → Unit Testing

Sprint Cycle

A sprint is a short fixed period (1-4 weeks) during which a scrum team work to complete a set of planned tasks and deliver a usable product increment

Sprint Planning (Sprint Backlog, Tasks)
↓
Daily Scrum Meeting (What to do today, and next)
↓
Sprint Review (Presentation, PBIs Don, Feedbacks)
↓
Sprint Retrospective (What went well, improvements, Actions)

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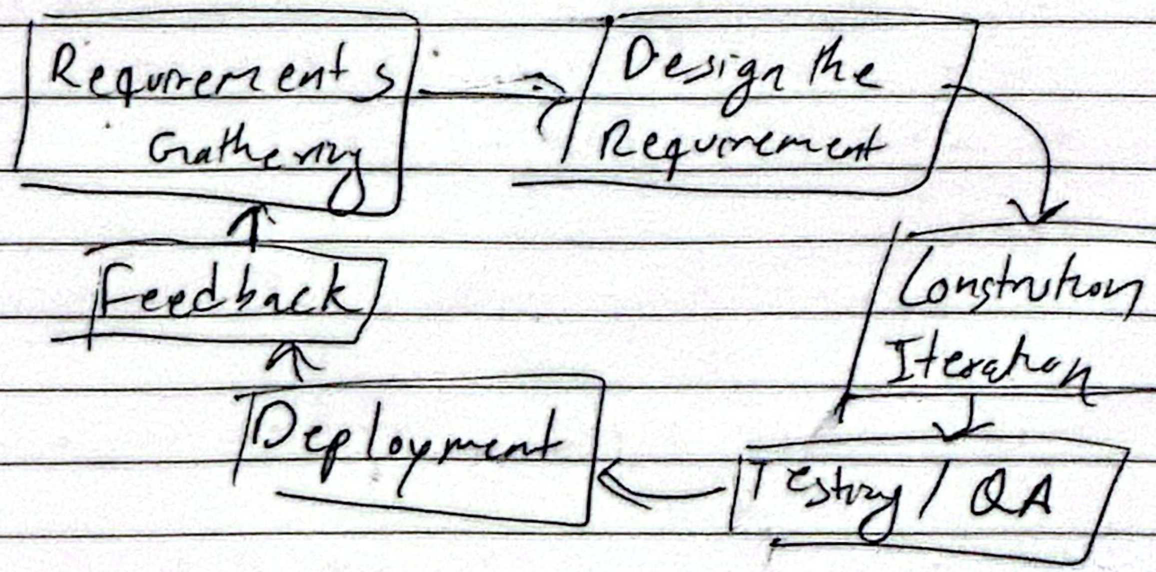
Ittrivtive SDLC Models

Agile/Scrum model

The Agile Model is an iterative and incremental approach designed to delive software quickly while adapting to changing customer requirements through continuous collaboration and feedback.

- Adapts quickly to changing customer req.
- Focuses on rapid and continuous software deli
- Encourages continuous customer feedback & collaboration
- Reduces time and effort by eliminating unnecessa tasks.

Agile SDLC



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For eg :- Student learning App.
Sprint 1: → Login. Feature → Unit → Integration → System → Acceptance

Sprint 2 → Course Search → Unit → Integration → System → Acceptance

Sprint 3 → Registration → Unit → Integration → → System → Acceptance.

:

Level of Testing

Test level are different stages of test used to find defects at different part of a software system

i) Unit Testing
Test individual functions.

ii) Integration testing
Test whether different component work together correctly

iii) System Testing
Test the complete system

iv) Acceptance Testing
Check wheathe the software meets user req